

Torsion transplantation and a uniform reverse Kohler–Jobin inequality for Robin problems

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Abstract

We prove that balls maximize the product of the first Robin eigenvalue and a sufficiently large power of the torsional rigidity among bounded planar Lipschitz sets of prescribed area. The exponent threshold is independent of the positive Robin parameter and the area. This answers the planar case of the reverse Kohler–Jobin question posed by Buttazzo, Cito and Solombrino. More generally, the argument applies to the Robin p -Laplacian in dimension n when $1 < p \leq n/(n-1)$, using the integral of the p -torsion function. The main step is a composition identity with a nonpositive boundary correction, which transplants the ball eigenfunction through the torsion function and preserves the reference energy. The known pointwise Talenti comparison then yields a linear upper bound for the spectral deficit in terms of the torsional deficit. An elementary global bound completes the optimization argument. For larger exponents, the product deficit is equivalent to the torsional deficit, uniformly in the Robin parameter. The higher-dimensional linear Robin problem is not resolved.

1. Introduction and main results

For a fixed positive Robin parameter, the ball minimizes the first eigenvalue and maximizes torsional rigidity among sets of prescribed volume. These two inequalities act in opposite directions on their product. Increasing the power of torsion suggests that its maximizer should eventually maximize the product, but the separate isoperimetric inequalities do not prove this assertion: one needs quantitative control of the competing increase in the eigenvalue.

Buttazzo, Cito and Solombrino [BCS, Section 5, Q2] ask whether the ball maximizes the first Robin eigenvalue times a sufficiently large power of Robin torsional rigidity. The corresponding Dirichlet reverse Kohler–Jobin inequality is established by Bucur, Lamboley, Nahon and Prunier [BLNP, Corollary 1.6]. Here we answer the Robin question in the plane. Our exponent threshold is uniform in both the positive boundary parameter and the prescribed area. The proof also gives a nonlinear extension in the range of the known pointwise Robin rearrangement theorem.

Let $n \geq 2$, $p > 1$ and $\beta > 0$. For a nonempty bounded open set $\Omega \subset \mathbb{R}^n$ with Lipschitz boundary, define

$$\begin{aligned} \mathcal{E}_{\beta,p,\Omega}(f) &= \int_{\Omega} |\nabla f|^p dx + \beta \int_{\partial\Omega} |f|^p d\mathcal{H}^{n-1}, \\ \lambda_{\beta,p}(\Omega) &= \inf_{f \in W^{1,p}(\Omega) \setminus \{0\}} \frac{\mathcal{E}_{\beta,p,\Omega}(f)}{\int_{\Omega} |f|^p dx}. \end{aligned} \tag{1.1}$$

Let $w = w_{\beta,p,\Omega}$ be the nonnegative weak solution of

$$-\Delta_p w = 1 \quad \text{in } \Omega, \quad |\nabla w|^{p-2} \partial_\nu w + \beta w^{p-1} = 0 \quad \text{on } \partial\Omega, \quad (1.2)$$

where $\Delta_p w = \operatorname{div}(|\nabla w|^{p-2} \nabla w)$, and put

$$\tau_{\beta,p}(\Omega) = \int_{\Omega} w \, dx. \quad (1.3)$$

Our convention for τ is the integral of w , rather than its $(p-1)$ -st power. For $p=2$ it is the usual torsional rigidity T_β . Let B be a ball with $|B| = |\Omega| = V$. We omit subscripts when the parameters are fixed and write

$$\theta(\Omega) = \frac{\tau(\Omega)}{\tau(B)}, \quad \mathcal{R}_q(\Omega) = \frac{\lambda(\Omega)\tau(\Omega)^q}{\lambda(B)\tau(B)^q}. \quad (1.4)$$

Theorem 1.1 (uniform reverse Robin inequality). Fix $n \geq 2$ and $1 < p \leq n/(n-1)$. There is a finite number $Q(n,p) > \max\{1, p-1\}$ such that, for every $\beta > 0$, $V > 0$ and nonempty bounded open Lipschitz set Ω of volume V ,

$$\lambda_{\beta,p}(\Omega)\tau_{\beta,p}(\Omega)^q \leq \lambda_{\beta,p}(B)\tau_{\beta,p}(B)^q \quad (q \geq Q(n,p)). \quad (1.5)$$

For every such q ,

$$1 - \theta(\Omega) \leq 1 - \mathcal{R}_q(\Omega) \leq q(1 - \theta(\Omega)). \quad (1.6)$$

Equality in (1.5) holds if and only if Ω is a translate of B .

In particular, $Q(2,2)$ supplies a single exponent threshold for the planar problem [BCS, Q2], simultaneously for all positive Robin parameters. Neither convexity nor connectedness is required. Formula (1.6) says that, for a fixed admissible exponent, the normalized product deficit and torsional deficit are equivalent with constants independent of β , V and Ω .

The estimate underlying the theorem is the following one-sided spectral control.

Theorem 1.2 (linear spectral control). Under the hypotheses of Theorem 1.1, there is $C(n,p) < \infty$, independent of β and V , such that

$$0 \leq \frac{\lambda(\Omega)}{\lambda(B)} - 1 \leq C(n,p)(1 - \theta(\Omega)) \quad \text{if } 1 - \theta(\Omega) \leq C(n,p)^{-1}. \quad (1.7)$$

The essential rearrangement input is [AGM, Theorem 1.2(i)]:

$$w_{\Omega}^{\#} \leq w_B \quad \text{a.e. on } B \quad \left(1 < p \leq \frac{n}{n-1}\right), \quad (1.8)$$

where $\#$ denotes Schwarz decreasing rearrangement. For $n = p = 2$, this is the comparison of Alvino, Nitsch and Trombetti [ANT]. We use their theorem with constant forcing and exactly the same Robin parameter on both sets. The only dimensional restriction in our proof occurs at (1.8).

We also use the Robin p -Faber-Krahn inequality and rigidity [BD, Theorem 1.1]. Its extension from connected domains to the present class is elementary. A bounded Lipschitz set has finitely many components Ω_j , each Lipschitz, and its first eigenvalue is $\min_j \lambda(\Omega_j)$. The eigenvalue of a ball strictly decreases with its radius at fixed β : dilation by $t > 1$ changes a trial quotient with

gradient part A and boundary part D into $t^{-p}A + t^{-1}D$, which is at most $t^{-1}(A + D)$. Consequently $\lambda(tB) \leq t^{-1}\lambda(B) < \lambda(B)$. Apply [BD] to every component and then enlarge its comparison ball to volume V . This proves $\lambda(\Omega) \geq \lambda(B)$, with strict inequality if Ω is disconnected; the connected equality case is precisely [BD]. The positivity and simplicity of the ball ground state are standard parts of this Robin eigenvalue theory.

2. A composition identity with a favorable boundary term

Lemma 2.1. Suppose w is a bounded nonnegative weak Robin p -torsion function on a bounded Lipschitz set, with $0 \leq w \leq M$. Let $h : [0, M] \rightarrow [0, \infty)$ be C^1 , with $h(0) = 0$, and set

$$H(s) = \int_0^s |h'(t)|^p dt, \quad G(s) = h(s)^p - s^{p-1}H(s). \quad (2.1)$$

Then $h(w) \in W^{1,p}(\Omega)$ and

$$\mathcal{E}_{\beta,p,\Omega}(h(w)) = \int_{\Omega} H(w) dx + \beta \int_{\partial\Omega} G(w) d\mathcal{H}^{n-1} \leq \int_{\Omega} H(w) dx. \quad (2.2)$$

Proof. Extend h and H as Lipschitz functions beyond their compact intervals. Sobolev composition and the trace identity justify testing (1.2) with $H(w)$. Since $H' = |h'|^p$, this gives

$$\int_{\Omega} |\nabla h(w)|^p dx = \int_{\Omega} H(w) dx - \beta \int_{\partial\Omega} w^{p-1} H(w) d\mathcal{H}^{n-1}.$$

Adding the Robin energy of the trace gives the identity. Hölder's inequality yields

$$h(s)^p = \left| \int_0^s h'(t) dt \right|^p \leq s^{p-1} \int_0^s |h'(t)|^p dt,$$

so $G \leq 0$. The interior bounds on w hold for its Sobolev trace as well: apply the trace operator to the truncations $(w - M)_+$ and w_- . Thus all boundary terms above are integrable. This proves (2.2). $\square$

If h is linear on $[0, b]$, then $G(b) = 0$. We will impose precisely this extension below the boundary value of the ball torsion. It makes the boundary inequality an equality on the reference ball, without requiring the trace of the competing torsion function to be constant.

3. Encoding the ball eigenfunction

Let $B = B_R$ and write $p' = p/(p-1)$. The radial torsion function $v = w_B$ is

$$v(r) = b + \frac{p-1}{p} n^{-1/(p-1)} (R^{p'} - r^{p'}), \quad b = \left(\frac{R}{n\beta} \right)^{1/(p-1)}. \quad (3.1)$$

Let ϕ be the positive first eigenfunction on B , normalized by $\int_B \phi^p dx = 1$, and write $\lambda_B = \lambda(B)$. Simplicity and rotational invariance make ϕ radial. Integrating its radial equation gives

$$-\phi'(r) = \left[\lambda_B r^{1-n} \int_0^r s^{n-1} \phi(s)^{p-1} ds \right]^{1/(p-1)}. \quad (3.2)$$

In particular, ϕ strictly decreases for $r > 0$. Put $M = v(0)$ and define

$$h(v(r)) = \phi(r) \quad (0 \leq r \leq R), \quad h(s) = \frac{\phi(R)}{b}s \quad (0 \leq s \leq b). \quad (3.3)$$

For $0 < r \leq R$,

$$h'(v(r)) = \left[n\lambda_B r^{-n} \int_0^r s^{n-1} \phi(s)^{p-1} ds \right]^{1/(p-1)}. \quad (3.4)$$

The derivative has the finite positive limit $\lambda_B^{1/(p-1)}\phi(0)$ at the center. At $r = R$, the two Robin boundary conditions give $h'(b) = \phi(R)/b$, so the pieces in (3.3) join in C^1 . The limiting derivative also verifies differentiability at M , for example by the mean value theorem. Thus h is increasing and C^1 on $[0, M]$.

Its derivative is nondecreasing in its argument. Indeed, the radial average in (3.4) decreases with r , since ϕ decreases, while $v(r)$ decreases. Therefore both h and h' take their maxima at M , and

$$L := \max_{0 \leq s \leq M} (h(s)^p)' = p\lambda_B^{1/(p-1)}\phi(0)^p. \quad (3.5)$$

For the function H in Lemma 2.1, the linear extension makes $G(b) = 0$. Applying the identity on B gives

$$\int_B H(v) dx = \mathcal{E}_{\beta,p,B}(\phi) = \lambda_B, \quad \int_B h(v)^p dx = 1. \quad (3.6)$$

Only first derivatives of h are used, including at the center of the ball.

4. From torsional deficit to global product optimality

This section assumes (1.8). Define

$$\delta = \tau(B) - \tau(\Omega), \quad c = L\tau(B), \quad \theta = \frac{\tau(\Omega)}{\tau(B)}. \quad (4.1)$$

The comparison ensures $0 < \theta \leq 1$ and bounds w by M . Lemma 2.1, equimeasurability and the monotonicity of H imply

$$\mathcal{E}_{\beta,p,\Omega}(h(w)) \leq \int_{\Omega} H(w) dx = \int_B H(w^\#) dx \leq \int_B H(v) dx = \lambda_B. \quad (4.2)$$

The denominator satisfies

$$\begin{aligned} 0 \leq 1 - \int_{\Omega} h(w)^p dx &= \int_B (h(v)^p - h(w^\#)^p) dx \\ &\leq L \int_B (v - w^\#) dx = L\delta. \end{aligned} \quad (4.3)$$

The trial function is nonzero, since h is strictly increasing and w is nonzero and nonnegative. Its Rayleigh quotient yields

$$\frac{\lambda(\Omega)}{\lambda_B} \leq \frac{1}{1 - c(1 - \theta)} \quad \text{if } c(1 - \theta) < 1. \quad (4.4)$$

In particular, Faber-Krahn and (4.4) imply

$$0 \leq \frac{\lambda(\Omega)}{\lambda_B} - 1 \leq 2c(1 - \theta) \quad \text{if } 1 - \theta \leq \frac{1}{2c}. \quad (4.5)$$

Since $h(0) = 0$ and h^p has Lipschitz constant L , (3.6) gives $1 \leq L\tau(B)$, hence $c \geq 1$.

We also need a global bound. Testing the torsion equation with w gives $\mathcal{E}(w) = \tau(\Omega)$. Using w in the Rayleigh quotient and then applying Hölder gives

$$\lambda(\Omega) \leq \frac{\tau(\Omega)}{\int_{\Omega} w^p dx} \leq \frac{V^{p-1}}{\tau(\Omega)^{p-1}}. \quad (4.6)$$

This estimate alone does not identify the maximizing ball, but it controls domains outside the neighborhood covered by (4.5).

Proposition 4.1 (a threshold from ball data). Set

$$\begin{aligned} k &= \frac{\lambda_B \tau(B)^{p-1}}{V^{p-1}}, & t_0 &= 1 - \frac{1}{2c}, \\ q_0 &= \max \left\{ 2c, p - 1 + \frac{\log(1/k)}{-\log t_0} \right\}. \end{aligned} \quad (4.7)$$

Then $\mathcal{R}_q(\Omega) \leq 1$ for all $q \geq q_0$. For all $q \geq q_0 + 1$,

$$1 - \theta \leq 1 - \mathcal{R}_q(\Omega) \leq q(1 - \theta). \quad (4.8)$$

Proof. By (4.6), $0 < k \leq 1$, and $1/2 \leq t_0 < 1$. If $\theta \geq t_0$, then (4.4) gives

$$\log \frac{\lambda(\Omega)}{\lambda_B} \leq -\log(1 - c(1 - \theta)) \leq 2c(1 - \theta) \leq -2c \log \theta.$$

Thus $\lambda(\Omega)/\lambda_B \leq \theta^{-2c}$, proving the product bound for $q \geq q_0$. If $\theta \leq t_0$, (4.6) gives instead

$$\mathcal{R}_q(\Omega) \leq k^{-1} \theta^{q-(p-1)} \leq k^{-1} t_0^{q-(p-1)} \leq 1 \quad (q \geq q_0),$$

by the definition of q_0 . This covers every $\theta \in (0, 1]$.

For $q \geq q_0 + 1$, apply the product bound at q_0 and use Faber-Krahn to obtain

$$\theta^q \leq \mathcal{R}_q(\Omega) = \mathcal{R}_{q_0}(\Omega) \theta^{q-q_0} \leq \theta^{q-q_0} \leq \theta.$$

Bernoulli's inequality $1 - \theta^q \leq q(1 - \theta)$ proves (4.8). $\square$

5. Uniformity in the Robin parameter

The constants in Proposition 4.1 involve only the reference ball. The next step shows that their dependence on its radius and on β can be eliminated.

Proposition 5.1. For fixed $n \geq 2$ and $p > 1$, the quantities c and k in (4.1) and (4.7), computed on a ball with Robin parameter $\beta > 0$, satisfy

$$1 \leq c \leq C_0(n, p) < \infty, \quad 0 < k_0(n, p) \leq k \leq 1. \quad (5.1)$$

Consequently the supremum of q_0 over all radii and all positive Robin parameters is finite.

Proof. Under $x = Ry$, the dimensionless boundary parameter is $\alpha = \beta R^{p-1}$. The eigenvalue scales as R^{-p} , the normalized eigenfunction as $R^{-n/p}$, and τ as $R^{n+p'}$. It follows directly from (3.5), (4.1) and (4.7) that c and k are unchanged by this simultaneous rescaling. We may therefore take $R = 1$ and vary β in $(0, \infty)$.

Write $V = |B_1|$. Integrating (3.1) gives

$$\tau(B_1) = V \left[b + \frac{n^{-1/(p-1)}}{n + p'} \right], \quad b = (n\beta)^{-1/(p-1)}. \quad (5.2)$$

Let λ_D be the first Dirichlet p -eigenvalue on B_1 . The Rayleigh principle, using a Dirichlet eigenfunction and a constant function respectively, gives

$$0 < \lambda_B \leq \lambda_D, \quad \lambda_B \leq n\beta. \quad (5.3)$$

Equation (3.2) implies

$$0 \leq -\phi'(r) \leq \lambda_D^{1/(p-1)} \phi(0) r^{1/(p-1)}. \quad (5.4)$$

Choose $r_* \in (0, 1)$ depending only on n, p so that integration of (5.4) yields $\phi(r) \geq \phi(0)/2$ on $[0, r_*]$. Normalization then bounds $\phi(0)$ uniformly above, while monotonicity gives $\phi(0) \geq V^{-1/p}$. The normalized radial eigenfunctions are therefore uniformly bounded and equicontinuous on $[0, 1]$.

As $\beta \downarrow 0$, (5.3) and (3.2) show that the oscillation of ϕ tends to zero. Normalization implies uniform convergence to $V^{-1/p}$. The integrated eigenfunction equation is

$$\lambda_B \int_{B_1} \phi^{p-1} dx = \beta |\partial B_1| \phi(1)^{p-1}. \quad (5.5)$$

Hence $\lambda_B/(n\beta) \rightarrow 1$. Inserting this limit and (5.2) into the definitions of c and k gives

$$c \longrightarrow p, \quad k \longrightarrow 1 \quad (\beta \downarrow 0). \quad (5.6)$$

For the other endpoint, (5.4) and the Robin condition imply $\phi(1) = O(\beta^{-1/(p-1)})$. Given any sequence $\beta \rightarrow \infty$, take a subsequence along which λ_B converges to λ_* and ϕ converges uniformly to ϕ_* . The limit is nonnegative, normalized, and vanishes at $r = 1$. Equation (3.2) passes to the limit. Its right-hand side is bounded by a constant times $r^{1/(p-1)}$, so dominated convergence also gives convergence of the gradient energies. Furthermore,

$$\beta \int_{\partial B_1} \phi^p d\mathcal{H}^{n-1} = O(\beta^{-1/(p-1)}) \longrightarrow 0.$$

Therefore $\phi_* \in W_0^{1,p}(B_1)$ and

$$\lambda_D \leq \int_{B_1} |\nabla \phi_*|^p dx = \lambda_* \leq \lambda_D.$$

Thus $\lambda_* = \lambda_D$ and ϕ_* is the normalized nonnegative first Dirichlet eigenfunction ϕ_D . Simplicity identifies every subsequential limit, so the whole family converges. Writing $\tau_D = \int_{B_1} w_D dx$, we obtain

$$\begin{aligned} c &\longrightarrow p\lambda_D^{1/(p-1)}\phi_D(0)^p\tau_D < \infty, \\ k &\longrightarrow \frac{\lambda_D\tau_D^{p-1}}{V^{p-1}} > 0 \quad (\beta \rightarrow \infty). \end{aligned} \tag{5.7}$$

For completeness, the ball data are continuous at every $\beta_* \in (0, \infty)$. The same radial compactness and gradient convergence apply along $\beta \rightarrow \beta_*$. The limiting normalized trial gives $\lambda_* \geq \lambda_{\beta_*, p}(B_1)$. Conversely, the fixed ground state at β_* is an admissible trial for nearby parameters and gives the reverse upper limit. Simplicity again identifies the uniform eigenfunction limit. Thus λ_B and $\phi(0)$, and hence c and k , are continuous.

The endpoint limits and continuity prove (5.1), since $c \geq 1$ and $k \leq 1$ were proved in Section 4. The denominator $-\log(1 - 1/(2c))$ in (4.7) is bounded below by the positive constant $-\log(1 - 1/(2C_0))$. Hence q_0 has a finite supremum. $\square$

Proof of Theorems 1.1 and 1.2. Set

$$Q(n, p) = 1 + \sup_{\alpha > 0} q_0(n, p, \alpha), \quad C(n, p) = 2 \sup_{\alpha > 0} c(n, p, \alpha), \tag{5.8}$$

where the ball has radius one and boundary parameter α . Proposition 5.1 makes these constants finite. Equation (4.5) proves Theorem 1.2; Proposition 4.1 proves (1.5) and (1.6). If equality holds in (1.5), (1.6) gives $\theta = 1$. Equation (4.4) then yields $\lambda(\Omega) \leq \lambda(B)$; Faber-Krahn and its rigidity statement give that Ω is a translate of B . Such a translate plainly attains equality. $\square$

6. Consequences and remaining questions

Theorem 1.1 turns any available lower bound for the normalized torsional deficit into the same lower bound for the product deficit. For example, in the planar linear case the quantitative Robin Saint-Venant inequality gives a positive constant $a(V, \beta)$ such that

$$1 - \mathcal{R}_q(\Omega) \geq a(V, \beta)\mathcal{A}(\Omega)^2 \quad (q \geq Q(2, 2)), \tag{6.1}$$

where $\mathcal{A}(\Omega) = \inf_{x \in \mathbb{R}^2} |\Omega \triangle (B + x)|/V$ is Fraenkel asymmetry. This is a direct consequence of (1.6) and the known torsional remainder, as given for example in [ABCMP, Corollary 4.1]. The geometric estimate is a prior input; we do not claim that $a(V, \beta)$ can be chosen uniformly in β by that argument.

The proof of the product theorem uses no compactness of the class of competing domains and no boundary perturbation of the ball. The composition identity preserves the reference energy, and pointwise torsion comparison controls the loss in the trial denominator. The near-optimal estimate (4.4) and the global bound (4.6) are the two distinct ingredients needed to reach every domain.

The same composition construction with zero boundary value is compatible with the Dirichlet problem. The linear Dirichlet reverse inequality is already a theorem of [BLNP]; it is not an additional open-problem solution asserted here. In the Robin setting, the linear case $n \geq 3$ remains beyond the pointwise input (1.8). The known L^1 and L^p torsion comparisons do not by themselves control the nonlinear numerator $\int H(w)$ in (4.2). Establishing a suitable substitute, or a different trial construction, remains necessary to extend the present proof.

The exponent (5.8) is sufficient and is not asserted to be optimal. Determining the least admissible exponent, extending the range of p , and settling the forward Robin Kohler-Jobin question [BCS, Q1] remain separate problems.

References

[ABCMP] V. Amato, R. Barbato, S. Cito, A. L. Masiello and G. Paoli, *A quantitative Talenti-type comparison result with Robin boundary conditions*, preprint, [arXiv:2511.11316v1](https://arxiv.org/abs/2511.11316v1).

[AGM] V. Amato, A. Gentile and A. L. Masiello, *Comparison results for solutions to p -Laplace equations with Robin boundary conditions*, *Annali di Matematica Pura ed Applicata* **201** (2022), 1189-1212. DOI: [10.1007/s10231-021-01153-y](https://doi.org/10.1007/s10231-021-01153-y).

[ANT] A. Alvino, C. Nitsch and C. Trombetti, *A Talenti comparison result for solutions to elliptic problems with Robin boundary conditions*, *Communications on Pure and Applied Mathematics* **76** (2023), 585-603. DOI: [10.1002/cpa.22090](https://doi.org/10.1002/cpa.22090).

[BCS] G. Buttazzo, S. Cito and F. Solombrino, *Relations between principal eigenvalue and torsional rigidity with Robin boundary conditions*, *Milan Journal of Mathematics* **94** (2026), 369-386. DOI: [10.1007/s00032-026-00438-2](https://doi.org/10.1007/s00032-026-00438-2).

[BD] D. Bucur and D. Daners, *An alternative approach to the Faber-Krahn inequality for Robin problems*, *Calculus of Variations and Partial Differential Equations* **37** (2010), 75-86. DOI: [10.1007/s00526-009-0252-3](https://doi.org/10.1007/s00526-009-0252-3).

[BLNP] D. Bucur, J. Lamboley, M. Nahon and R. Prunier, *Sharp quantitative stability of the Dirichlet spectrum near the ball*, preprint, [arXiv:2304.10916v2](https://arxiv.org/abs/2304.10916v2).